

Drone Fire Control System (FCS) Architecture Specification

1. Executive Summary & Communication Strategy

This document specifies the revised hardware communication and functional allocation architecture for the Drone Fire Control System (FCS). Following the addition of a second STM32 microcontroller for motion control and thermal monitoring, the system utilizes a unified **CAN Bus** architecture by multi-drop network topology, avoiding the need for complex kernel-level modifications on the Jetson Orin Nano's second CAN channel.

Key Infrastructure Decisions:

- **CAN Bus Shared Network:** The secondary STM32 is wired in parallel to the existing active CAN Bus channel. Message separation is handled via unique software-defined CAN IDs (e.g., 0x100+ for safety/firing, 0x200+ for motion/telemetry). Terminal 120-ohm resistors are placed strictly at the two physical endpoints of the bus.
- **On-Drone Ethernet Topology:** Since the GCS Panasonic Tablet provides a single Ethernet port, a *Miniature Rugged Airborne Ethernet Switch* must be mounted on the drone. The Air Data Terminal (Radio Receiver) feeds into this switch, which splits the connection directly to the Jetson Orin Nano (control loop data) and the Micro300 Camera (live video stream).
- **Hierarchical Control (No Direct GCS-to-MCU Link):** Microcontrollers do not communicate directly with the ground station. The Jetson acts as the local system gateway. This ensures hard real-time safety loop isolation from RF jitter, latency, or packet drop.

2. Functional Allocation Matrix

System Component	Primary Subsystem Functions	Hardware Interfaces	Critical Fail-Safe Mechanism
GCS PC / Tablet (Panasonic Rugged)	Runs C# tracking application. Processes live video streams, handles object detection/aiming logic, and transmits target coordinates (X, Y) to the drone via Ethernet.	Ethernet (RF Link to Drone), Operator Controls.	If the tracking software or RF link crashes, coordinate transmission halts instantly, preventing unverified system operation.

■ FCS JETSON (Orin Nano)	Acts as Mission & Ballistic Computer. Receives X,Y inputs from GCS, integrates Laser Range Finder data, calculates real-time ballistic adjustments (Azimuth/Elevation corrections), transmits positioning data to the gimbal controller, and sends electronic lock validation to the safety controller.	Ethernet Switch, Laser Range Finder (LRF), CAN Bus (Shared channel).	If the Ethernet link to the GCS is lost, it immediately ceases motion commands and halts the 2-second Heartbeat (HB) signal sent to STM32 #1.
■■ STM32 #1 (FCS/ISM 1)	Handles rigid hardware firing logic and safety interlocks. Evaluates firing criteria and controls the firing circuit gates.	Physical Trigger, Recoil Sensor, Electronic Safety Gates (Safety 1 & 2).	Heartbeat Timeout: If the 2-second Heartbeat from the Jetson ceases, it immediately activates an electronic hardware lock to block firing.
■■ STM32 #2 (FCS/ISM 2)	Performs real-time motion control and environment monitoring. Executes fast PID loops for stabilization and drives positioning axis servos.	Azimuth & Elevation Servos, Angle Sensors, IMU, Temperature Sensor.	Thermal Protection: If the temperature exceeds 100°C, it triggers an autonomous hardware shutdown of the servos, locks positions, and alerts the Jetson.

3. Dual-Layer Safety Interlock Logic (Safety 1 & 2)

To meet rigorous safety requirements and avoid accidental weapon operation, weapon firing requires the simultaneous fulfillment of two distinct operational states:

- **Safety 1 — STANDBY State (System Authorization):** Opened automatically via CAN Bus command by the Jetson once a valid target lock is confirmed and target parameters match calculated ballistic profiles.
- **Safety 2 — ARM State (Operator Authorization):** Opened explicitly when the physical trigger signal from the GCS operator reaches the controller.

The physical weapon firing circuit is completed **ONLY** when Safety 1 AND Safety 2 are simultaneously active. If either condition fails or the Jetson heartbeat drops, the circuit breaks instantly.

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